MUSAWANDILE SHAMASE

Junior Industrial Automation Engineer | PLC Programming · Control Systems · Instrumentation

BEngTech Electronic Engineering | PLCDOJO PLC Fundamentals (Level I) Certificate

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PROFESSIONAL SUMMARY

Electronic Engineering graduate (BEngTech, Durban University of Technology, 2025) with a control systems and process instrumentation background and hands-on PLC programming in RSLogix 500. Holds the PLCDOJO PLC Fundamentals (Level I) certificate and has programmed and tested ladder-logic projects covering motor start/stop with seal-in, Hand/Off/Auto control, logic gates, and digital and analog tank-pump control, each documented on an online portfolio using the five-step PLC program development process. Adds embedded-systems experience in sensor interfacing, relay control and safety interlock logic. Seeking a junior industrial automation role to apply PLC programming and control knowledge in a working plant environment.

TECHNICAL SKILLS

Industrial Automation: PLC architecture, scan cycles, digital and analog I/O, ladder logic (RSLogix 500), on-delay timers, seal-in (latching) logic, Hand/Off/Auto control, analog scaling and comparison instructions (SCP, LES, GRT), subroutine-structured programs (JSR), I/O addressing, program documentation and emulator testing

Control Systems & Instrumentation: Control systems theory, process control, process instrumentation, sensors and signals, signals and systems

Embedded Systems & Electronics: Microcontroller programming (Arduino), sensor interfacing, relay control, hardware interrupts, analogue and digital electronics, circuit design and troubleshooting, soldering, wiring and prototyping, electrical safety awareness

Programming: Ladder Logic, C/C++, Embedded C, Arduino C++, Python

Tools & Testing: RSLogix 500 and emulator, MATLAB, Arduino IDE, Microsoft Office, oscilloscope, digital multimeter

RELEVANT ACADEMIC COURSEWORK (BENGTECH, DUT)

- Fundamentals of Control Systems 2B (79%); Control Systems 3A; Process Control Systems 3B
- Fundamentals of Instrumentation 2A; Process Instrumentation 3A
- Fundamentals of Microcontrollers 2A (83%); Embedded Systems 2B
- Fundamentals of Signals and Systems 2A (82%); Digital Electronics IB (86%); Fundamentals of Power Engineering 2A
- Engineering Ethics and Professional Skills 3B (76%)

PROJECTS

PLC Programming Projects | *RSLogix 500, ladder logic, tested in the RSLogix 500 emulator*

- Analog Tank–Pump Control: scaled one analog level sensor to 0–100% with SCP and compared it against 20% and 80% trip points (LES, GRT) to drive a seal-in fill mode and the pump and valve outputs.
- Digital Tank–Pump Control: controlled a pump and valve from low, low-low, high and high-high level switches, using two on-delay timers for fill and drain and a FILL_MODE bit.
- Blower Hand/Off/Auto (HOA) Control: implemented Hand, Off and Auto mode logic for a blower, documented with I/O tables and a tested expected-behaviour sequence.
- Motor Start/Stop with Seal-In: wired START (normally open) and STOP (normally closed) pushbuttons to a conveyor motor output, with a MAIN program calling separate digital I/O and control-logic subroutines.
- Logic Gates in Ladder Logic: built AND, OR, NOT, NAND, NOR, XOR and XNOR gates in an 8-rung program using internal bits.
- Each project is documented step by step (task, I/O, sequence of operation, program, test) at amwecodex.github.io.

Car Safety Interlock System | *Microcontrollers, Embedded C, safety logic*

- Developed an embedded interlock that prevents vehicle start-up until safety conditions, such as a fastened seatbelt, are confirmed.
- Configured hardware interrupts and wrote the Embedded C firmware that verifies safety conditions in real time, with attention to wiring integrity.

Smart Home Irrigation System | *Arduino, C++, soil moisture sensors, relay modules*

- Designed and prototyped an automated irrigation controller that reads soil moisture and switches a water pump through a relay.
- Wired the sensors, relay module and pump circuit, and wrote the C++ control program, including the irrigation interval logic.

Cable Fault Detection System | *Embedded systems, signal processing, sensors*

- Built a diagnostic system to detect and locate faults in electrical and signal cabling, integrating sensor interfaces and signal-processing logic.

PROFESSIONAL DEVELOPMENT & CERTIFICATIONS

- **PLC Fundamentals (Level I) Certificate**, PLCDOJO, issued October 2026: PLC architecture, scan cycles, digital and analog I/O, ladder logic, timers and process logic. Verify certificate
- **Industrial Automation Portfolio**: public write-ups of PLC projects with I/O tables, program structure, ladder-logic breakdowns and test results at amwecodex.github.io.

WORK EXPERIENCE

Merchandiser | *MyChina, Boksburg* **April 2026 – October 2026**

- Maintain product displays, replenish stock and keep inventory records accurate in a high-volume retail store.
- Follow store procedures and coordinate with colleagues to meet daily targets.

Educational Assistant | *KwaBulawayo High School (Department of Basic Education)* **June 2025 – November 2025**

- Supported teachers with lesson preparation, classroom administration and tracking of learner progress.
- Tutored high school learners in core STEM subjects.

Volunteer Programming Tutor | *Dataholics Student Organisation, Durban* **March 2022 – August 2024**

- Tutored undergraduate students in programming fundamentals, logic design and software debugging.
- Ran practical coding workshops and test-preparation sessions.

EDUCATION

Bachelor of Engineering Technology (BEngTech) in Electronic Engineering **Graduated 2025**

Durban University of Technology (DUT), Durban, South Africa

ADDITIONAL INFORMATION

- Code 10 (C1) driver's licence
- South African citizen, eligible to work immediately; open to relocating across South Africa
- References available on request